Durgakant Pushp

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Research Focus

Adaptive robotic intelligence through coupled search over embodiment, representation, and policy under constraints on information, computation, time, and energy. Research applications include autonomous racing, intelligent transportation systems, safe and adaptive autonomous driving, visual navigation, and heterogeneous robotic systems for field exploration and infrastructure inspection.

Education

Ph.D., Intelligent Systems Engineering **2021–2026**
Indiana University Bloomington

Dissertation: *Bounded Rational Robot: Embodiment, Representation and the Search for Intelligent Behavior*
Advisor: Prof. Lantao Liu

B.Tech., Computer Science and Engineering **2013–2017**
West Bengal University of Technology

Research and Professional Experience

Postdoctoral Research Fellow **2026–Present**
School of Mechanical Engineering, Purdue University; Purdue AI Racing (PAIR)

- Mentor a Ph.D. student on LiDAR-based localization for high-speed autonomous racing; guide research direction and technical development of the localization module.
- Contributed technical input across perception, planning, and control, and helped establish development goals and module priorities for PAIR's record-setting 2026 racing campaign.
- Contribute to autonomy-stack architecture and integration to support new research capabilities and future development.
- Extend IJRR research on visual decision-making toward controlled stopping during sensor failures in autonomous racing; simulation results obtained, with vehicle validation planned.
- Investigate capability diagnosis and embodiment adaptation: how an agent identifies physical limitations and evaluates feasible changes under resource constraints.

Team Lead, IU–Luddy Autonomous Racing Team **2023–2025**
Indiana University Bloomington

- Led research planning, task allocation, technical development, and system integration for a full-scale autonomous racing program.
- Directed development across localization, perception, planning, control, and data analysis for the Dallara AV-24 platform.
- Led and mentored a multidisciplinary team of doctoral, master's, undergraduate, and high-school students; coordinated development and on-track testing at IMS, LVMS, Laguna Seca, and Kentucky Motor Speedway.
- Team achieved 128 mph within one month of preparation, second place at the 2025 LVMS time trial with a 164 mph average, and fourth place at the 2025 Laguna Seca road-course event.
- Led racing-system development documented in the ISER autonomous racing stack publication; contributed sections to proposals supporting the program.

Research and Professional Experience (continued)

Graduate Research Assistant; Technical Project Lead **2023–2025**
Indiana University Bloomington; Army ERDC-supported robotic inspection of flood-control systems

- Led day-to-day research and project management for IU's work on robotic inspection of flood-control infrastructure, including research planning, and field experimentation.
- Developed a UAV–miniUGV deployable system for confined-space inspection and designed and field-tested MiniUGV$_2$, integrating mobility, manipulation, sensing, and mission autonomy.
- Worked within a multi-university collaboration involving TAMU, Rice, UT Austin, LSU, MSU, and IU; contributed proposal sections for the project.

Graduate Research Assistant; Technical Project Lead **2021–2023**
Indiana University Bloomington; ARL-supported Off-Road Autonomy / SARA Program

- Led day-to-day technical execution and managed research progress for IU's off-road autonomy work, including terrain perception and autonomous navigation.
- Developed autonomy modules transitioned into ARL's Robotics Enterprise software repositories; contributed proposal sections supporting the project.

Project Lead, AISUM Autonomous Systems Challenge **2021–2022**
Indiana University Bloomington; U.S. Navy CRANE / OUSD(R&E)

- Developed IU's single-member entry on GPS-denied exploration, selected as a national Top-8 finalist; prepared proposal sections and challenge reports.

Research Engineer; Field Experiment Lead **2017–2020**
TCS Research, Bangalore, India; collaboration with the Indian Institute of Science

- Developed multi-agent aerial and ground robotic systems for industrial automation, including autonomous navigation, aerial manipulation, object detection and tracking, and decentralized multi-robot coordination.
- Conducted research on aerial target detection, tracking, and autonomous interception, contributing to three peer-reviewed publications at IROS, ICUAS, and ICC.
- Led the development, system integration, and outdoor field experiments for MBZIRC 2020 Challenge 1, contributing to the team's Best Innovation Award.
- Contributed to robotic manipulation and adaptive gripper development, resulting in a granted Indian patent (IN 202321018297, 2024).

Research Mentoring and Scholarly Leadership

Selected contributions to student research and publication development.

- **LiDAR localization for high-speed racing, Purdue.** Mentor a Ph.D. student on the localization module and its research development.
- **Seeing Where to Deploy (IROS 2026).** Established the research direction and guided the first and second authors from project inception through manuscript preparation; contributed to study design and writing.
- **AFRDA (RA-L 2025).** Held weekly research meetings, helped shape the study through submission, and contributed to the field experiment and manuscript writing.
- **From Zero to High-Speed Racing (ISER 2025).** Led the underlying racing-system development and guided research execution and integration; collaborated with the first author, who led final manuscript completion.

Publications

My name is in bold. Preprints are listed separately.

Journal Articles

[J1] **D. Pushp**, W. Chen, Z. Chen, C. Luo, J. M. Gregory, and L. Liu. "Navigating the Wild: Pareto-Optimal Visual Decision-Making in Image Space." *The International Journal of Robotics Research (IJRR)*, 2025.
Paper of the Season, Q2 2025: Honorable Mention.

[J2] M. A. M. Khan, **D. Pushp**, and L. Liu. "AFRDA: Attentive Feature Refinement for Domain Adaptive Semantic Segmentation." *IEEE Robotics and Automation Letters (RA-L)*, 2025.

[J3] Z. Chen, **D. Pushp**, J. M. Gregory, and L. Liu. "Pseudo-Trilateral Adversarial Training for Domain Adaptive Traversability Prediction." *Autonomous Robots*, vol. 47, pp. 1155–1174, 2023.

Conference and Symposium Publications

[C1] H. Jardali, I. S. Mohamed, **D. Pushp**, and L. Liu. "Time-Varying Parameter MPC for Robust Path Tracking and Lateral Control in Full-Scale Autonomous Racing." *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[C2] S. Lee, S. M. Shithil, **D. Pushp**, L. Liu, and Z. Wang. "Seeing Where to Deploy: Metric RGB-Based Traversability Analysis for Aerial-to-Ground Hidden Space Inspection." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.

[C3] **D. Pushp**, S. Kalhapure, S. M. Shithil, and L. Liu. "MiniUGV$_2$: A Compact UAV-Deployable Tracked Ground Vehicle with Manipulation Capabilities." *International Symposium on Experimental Robotics (ISER)*, 2025.

[C4] H. Jardali, **D. Pushp**, Y. Yu, M. Ali, I. S. Mohamed, A. Murillo-Gonzalez, P. D. Coen, M. A.-M. Khan, R. C. Pulivendula, S. Park, L. Zhou, and L. Liu. "From Zero to High-Speed Racing: An Autonomous Racing Stack." *International Symposium on Experimental Robotics (ISER)*, 2025.

[C5] **D. Pushp**, J. Xu, Z. Chen, and L. Liu. "Context-Generative Default Policy for Bounded Rational Agent." *IEEE/RSJ IROS*, 2024.

[C6] M. Ali, **D. Pushp**, Z. Chen, and L. Liu. "Visual-Geometry Gaussian Process-based Navigable Space for Autonomous Navigation." *IEEE/RSJ IROS*, 2024.

[C7] **D. Pushp**, Z. Chen, C. Luo, J. M. Gregory, and L. Liu. "POVNav: A Pareto-Optimal Mapless Visual Navigator." *ISER*, 2023.

[C8] **D. Pushp**, J. Xu, and L. Liu. "Coordination of Bounded Rational Drones through Informed Prior Policy." *IEEE/RSJ IROS*, 2023.

[C9] **D. Pushp**, S. Kalhapure, K. Das, and L. Liu. "UAV-miniUGV Hybrid System for Hidden Area Exploration and Manipulation." *IEEE/RSJ IROS*, pp. 1297–1304, 2022.

[C10] J. Xu, **D. Pushp**, K. Yin, and L. Liu. "Decision-Making Among Bounded Rational Agents." *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2022.

[C11] Z. Chen, **D. Pushp**, and L. Liu. "CALI: Coarse-to-Fine Alignments for Unsupervised Domain Adaptation of Traversability Prediction." *Robotics: Science and Systems (RSS)*, 2022.

[C12] U. Rai, **D. Pushp**, A. Choudhary, A. Kumar, B. Chaitanya, R. Lima, N. S. Abhinay, R. Majumder, S. Jana, K. Das, and D. Ghose. "PARA-FM: Graphical User Interface for Emergency Flood Planning and Management." *Proceedings of the 5th World Congress on Disaster Management*, vol. II, pp. 100–106, Routledge, 2022.

Publications (continued)

[C13] R. Srivastava, R. Lima, C. Shinde, **D. Pushp**, and K. Das. "Estimation and Control for Autonomous UAV System to Neutralize Unknown Aerial Maneuvering Targets." *7th Indian Control Conference (ICC)*, pp. 117–122, 2021.

[C14] V. Anand, **D. Pushp**, R. Raj, and K. Das. "Gaussian Mixture Model (GMM)-Based Object Detection and Tracking Using Dynamic Patch Estimation." *IEEE/RSJ IROS*, pp. 4474–4481, 2019.

[C15] V. Anand, **D. Pushp**, R. Raj, and K. Das. "Gaussian Mixture Model (GMM)-Based Dynamic Object Detection and Tracking." *International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 1365–1371, 2019.

Preprint

M. Ali, H. Jardali, Y. Yu, **D. Pushp**, and L. Liu. "Minimalistic Autonomous Stack for High-Speed Time-Trial Racing." arXiv:2509.19636, 2025.

Other Conference Contribution

D. Pushp and S. Samaddar. "A Randomly Guided Algorithm for Exact Pattern Matching." National Conference on Recent Advances in Science and Technology (NCRAST), Heritage Institute of Technology, Kolkata, September 27–29, 2016.

Honors and Awards

2026 IJRR Paper of the Season Honorable Mention, Q2 2025.

2024 Second Place, Cheng Wu Innovation Challenge; single-member team.

2023 Audience Choice Award, MassRobotics Form & Function Challenge; single-member team.

2021–2022 Top-8 National Finalist, AISUM Challenge; single-member team.

2014, 2015, 2017 Boeing-sponsored RC aircraft competition awards, IIT Kharagpur.

2015, 2016 First Prize, RC Aircraft Flying Competition, NIT Jamshedpur.

2016 Globsyn Faculty Choice Award for analysis of aircraft crash data.

Team Competition Results

2026 PAIR: First Place, Time Trial; Second Place, Passing Competition, Indy Autonomous Challenge, Laguna Seca; record-setting lap.

2025 IU–Luddy: Second Place, LVMS Time Trial; Fourth Place, Laguna Seca Road Course, Indy Autonomous Challenge.

2020 Best Innovation Award & $100K Milestone Grant, Mohamed Bin Zayed International Robotics Challenge

Ranked in the top 3 globally during the Phase-1 technical evaluation; served as the lead developer for the core codebase that secured the funding.

Teaching Experience

Guest Lecturer, Autonomous Robots **August 2024**
Intelligent Systems Engineering, Indiana University Bloomington

Undergraduate/graduate course; approximately 20 students.

- Delivered a lecture on autonomous racing, covering full-stack autonomy, GNSS-based localization, PID and Pure Pursuit control, and an introduction to MPPI.

Teaching Assistant, Introduction to Cyber-Physical Systems **January–May 2024**
Intelligent Systems Engineering, Indiana University Bloomington

Undergraduate course; approximately 30 students.

- Supervised student projects using ROS and autonomy stacks, held office hours, and supported assessment.

Workshop Organizer and Instructor, Aeromodelling **September 2017**
Jadavpur University, Kolkata, India

Two-day workshop; approximately 250 undergraduate participants.

- Designed and delivered lectures on aerodynamics and flight mechanics; guided hands-on construction and flight of RC aircraft models.

Patent Applications

- R. Lima, **D. Pushp**, et al. "Robot-Mountable Adaptive Gripper Assembly for Intercepting Aerial Targets." Indian patent application No. 202321018297.
- **D. Pushp**, S. Kalhapure, and L. Liu. "UAV Tethered to UGV System." U.S. provisional patent application No. 63/914,830.

Professional Service and Outreach

- Peer reviewer: ICRA, IROS, IEEE Robotics and Automation Letters, and ISER.
- Invited reviewer, IEEE Intelligent Vehicles Symposium (IV), 2026.
- Research and robotics mentoring for graduate, undergraduate, and high-school students through autonomous systems projects.

Technical Expertise

Autonomy: localization; visual navigation; terrain perception; planning and control; multi-robot coordination; field testing and systems integration.

Programming and robotics: C++, Python; ROS 1/2, Gazebo, AirSim, MuJoCo.

Learning and vision: PyTorch, TensorFlow, OpenCV, scikit-learn.

Sensing: LiDAR, radar, RTK-GNSS, IMUs, wheel encoders, depth and localization cameras, event-based cameras, sonar, and proximity sensors.